Fluidization of Granular Media:

Principles, Experimental Study and Numerical Modelling

TIPE 2025 — Theme: *Transformation, transition, conversion*

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Abstract

This report presents a comprehensive study of the fluidization of granular media, carried out as part of the TIPE 2025 project whose theme is *Transformation, transition, conversion.* Fluidization refers to the transition of a bed of solid particles from a fixed state to a fluid-like behaviour, under the effect of an upward gas flow. This phenomenon is ubiquitous in industry: waste incinerators, cooling systems, chemical reactors, etc.

Our approach is structured around four main axes: **(1)** setting up an experimental apparatus and calibrating the measurement tools, **(2)** experimentally verifying the Darcy and Ergun laws governing flow through porous media, **(3)** developing a numerical Discrete Element Model (DEM), and **(4)** validating the model by comparison with experiments on five different granular media.

Contents

1 Introduction and Motivations

1.1 Industrial context

The fluidization of granular media is a fundamental technology in many industrial sectors. Its principle: a bed of solid particles is traversed by a fluid current (gas or liquid) fast enough for the particles to become suspended and adopt a collective behaviour similar to that of a liquid.

Among the concrete applications, one can mention:

- **Fluidized bed incinerators**: combustion air is injected from the bottom into a bed of sand or ash, ensuring homogeneous mixing and efficient combustion of waste.
- **Fluidized bed cooling systems**: for example, cooling hot alumina in aluminium production, where air fluidizes the alumina powder while exchanging heat with a cold water circuit.
- **Catalytic reactors** in petrochemistry (FCC — Fluid Catalytic Cracking process), **drying of pharmaceutical or food powders**, or **pneumatic conveying** of granular materials.

The advantage of the fluidized bed lies in its remarkable properties: very high heat transfer coefficient (comparable to that of an oil bath), temperature homogeneity, ease of handling and transport of the suspended solid.

1.2 Problem statement and objectives

Fluidization constitutes a *mechanical phase transition*: a granular medium shifts from a solid state (fixed bed) to a fluid state (fluidized bed) when the gas velocity exceeds a critical threshold $V_{\text{fluidization}}$. This phenomenon perfectly illustrates the 2025 theme *Transformation, transition, conversion.*

The objectives of this work are:

1. To observe and experimentally characterise the fluidization phenomenon.
2. To verify the theoretical physical laws governing it (Darcy, Ergun).
3. To develop a discrete element numerical model capable of reproducing the behaviour of a fluidized bed.
4. To validate this model on several granular media with different properties.

1.3 Physical principle of fluidization

Consider a vertical tube containing a bed of spherical particles, traversed by an upward air flow. At low flow rates, the bed remains fixed: the particles do not move, and the pressure drop ΔP between the inlet and outlet of the bed increases with the fluid velocity V, in accordance with Ergun's law.

When the fluid velocity exceeds the **minimum fluidization velocity** V_{mf}, the drag exerted by the fluid on the particles exactly compensates the apparent weight of the entire bed. The particles lift and start moving: the bed is *fluidized*. The pressure drop then reaches a constant plateau, equal to the apparent weight of the bed per unit cross-sectional area ($\Delta P_{\mathrm{plateau}} = M_{\mathrm{bed}} g/S$).

2 Setting Up the Experimental Apparatus

2.1 Description of the initial setup

The experimental apparatus was designed to recreate the conditions of a fluidized bed in the laboratory. It comprises the following elements:

- A **vertical tube** containing the particle bed, surrounded by a porous wall (fabric) allowing homogeneous distribution of air at the inlet.
- An **air compressor**, controlled by a **voltage regulator**, allowing the injected air flow rate to be varied.
- A **multimeter** to measure the voltage U across the compressor.
- A **differential manometer** (initial setup) to measure the pressure drop ΔP across the bed.
- A **pressure cooker** (used for the calibration of the flow rate caracteristic) to measure the flow rate and find its relation with U.

2.2 Problems encountered and solutions

2.2.1 Problem 1: Accuracy of the differential manometer

The initial differential manometer had two major limitations:

- **High uncertainty** in measurements (coarse graduation).
- **Limited range** at low pressures ($\Delta h < 30\,\mathrm{cm}$ of water, i.e. approximately $3\,\mathrm{kPa}$).

Solution: The differential manometer was replaced by the barometric sensor of a smartphone, used via the **Phyphox** application. Pressure measurements are made inside a sealed **pressure cooker** (volume $V = 5.5\,\mathrm{L}$), acting as a pressurised measurement chamber. The precision achieved is of the order of **one pascal**, and measurements are no longer limited in amplitude.

Figure 1: Photo of the experimental setup used until the end of the project (with unplugged manometer and sealed pressure cooker).

2.2.2 Problem 2: Determination of the flow rate–voltage characteristic

The voltage regulator controls the compressor, but the relationship between the voltage U and the volumetric flow rate Q_v is not known a priori. This characteristic therefore had to be established experimentally.

2.2.3 Problem 3: Overpressure component due to the porous wall

The porous wall surrounding the tube itself generates a pressure drop ΔP_{wall} that superimposes on the bed pressure drop. This contribution must be measured separately so it can be subtracted from raw measurements.

2.3 Calibration of the flow rate–voltage characteristic

2.3.1 Principle of the method

The measurement of the compressor volumetric flow rate Q_v relies on the ideal gas law. The empty pressure cooker is sealed and air is injected at a constant flow rate. The pressure inside the cooker increases linearly with time according to:

Pressurization equation of the pressure cooker

$$P(t) = P_0 + Q_v \frac{\varrho_{\text{air}}}{M_{\text{air}}} \frac{RT}{V} t \tag{2.1}$$

where P_0 is the initial pressure, $\varrho_{\text{air}} = 1.2\,\text{kg}\,\text{m}^{-3}$ is the density of air, $M_{\text{air}} = 29\,\text{g}\,\text{mol}^{-1}$ its molar mass, $R = 8.314\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$ the ideal gas constant, $T = 298\,\text{K}$ the ambient temperature, and $V = 5.5 \times 10^{-3}\,\text{m}^3$ the volume of the cooker.

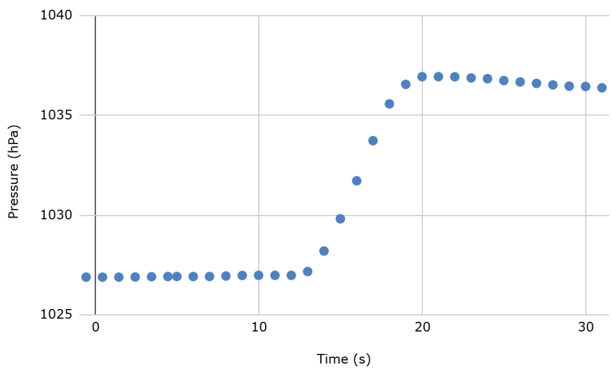


Figure 2: Typical pressure rise recorded in the pressure cooker (Phyphox sensor) as a function of time, for one voltage setting of the regulator. The linear transient regime (before the plateau) is used to extract the slope and compute the volumetric flow rate Q_v via equation (2.1).

2.3.2 Derivation of equation (2.1)

We perform a mass balance on the pressure cooker. The cooker contains n moles of air at pressure P and temperature T. During dt, the compressor injects an additional $\frac{\varrho_{\text{air}}}{M_{\text{air}}} Q_v\, dt$ moles, while no matter escapes (sealed cooker). The change in mole number is therefore:

$$dn = \frac{\varrho_{\text{air}}}{M_{\text{air}}} Q_v\, dt \tag{2.2}$$

By the ideal gas law, $n = \frac{PV}{RT}$, so $dn = \frac{V}{RT}\, dP$. Equating:

$$\frac{V}{RT}\, dP = \frac{\varrho_{\text{air}}}{M_{\text{air}}} Q_v\, dt \tag{2.3}$$

Integrating (initial conditions: $P(0) = P_0$, $v(0) = v_0 = 0$ since the compressor has just started), we recover equation (2.1). The volumetric flow rate is read directly from the slope of the $P(t)$ line:

Volumetric flow rate from the slope

$$Q_v = \text{slope} \times \frac{M_{\text{air}}}{\varrho_{\text{air}}} \times \frac{V}{RT} \qquad (2.4)$$

2.3.3 Calibration results

For each voltage value U of the regulator (from 100 V to 185 V), the slope of the $P(t)$ curve is extracted by linear regression, and the flow rate Q_v is then calculated via equation (2.4). Figure 3 shows the family of pressure-rise slopes obtained for all voltage settings, while Figure 4 presents the resulting $Q_v = f(U)$ characteristic. The results show a **linear relationship** between Q_v and U over the range of flow rates of interest, which greatly simplifies use of the apparatus.

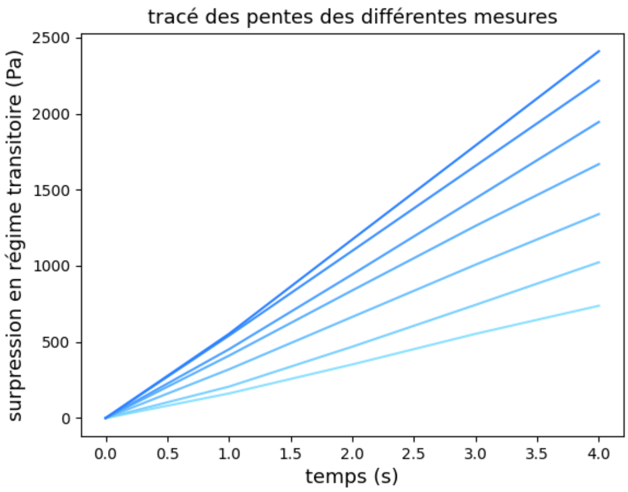


Figure 3: Family of pressure slopes $\Delta P(t)$ obtained during the transient calibration measurements, for the seven voltage settings of the regulator (from lightest to darkest blue as U increases). Each slope directly gives one value of Q_v through equation (2.4).

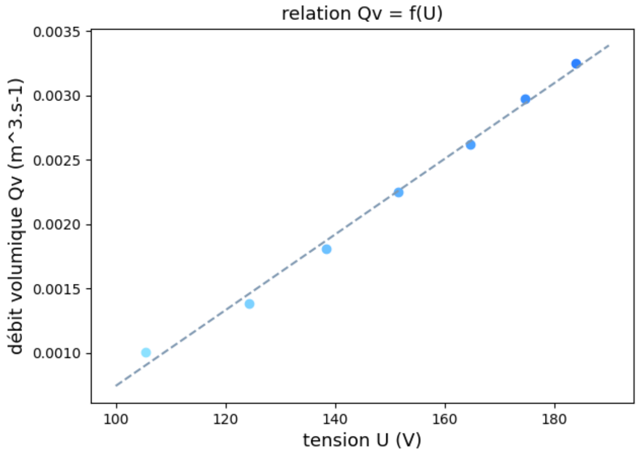


Figure 4: Volumetric flow rate Q_v as a function of regulator voltage U. The dashed line is a linear fit; the characteristic is linear over the full range of voltages used in the fluidization experiments (100–185 V).

The apparent fluid velocity in the tube of cross-sectional area $S = \pi(5.5\times10^{-2}/2)^2\ \mathrm{m}^2$ is then:

$$V = \frac{Q_v}{S} \tag{2.5}$$

2.4 Measurement of the overpressure component due to the porous wall

2.4.1 Experimental protocol

To measure $\Delta P_{\text{wall}}(V)$, air is circulated only through the porous wall, without any particle bed. The Phyphox application records the pressure in the cooker in real time. A Python script automatically detects the **pressure plateaus** corresponding to the different regulator voltages, by searching for time intervals where the derivative $|dP/dt|$ is below a threshold K.

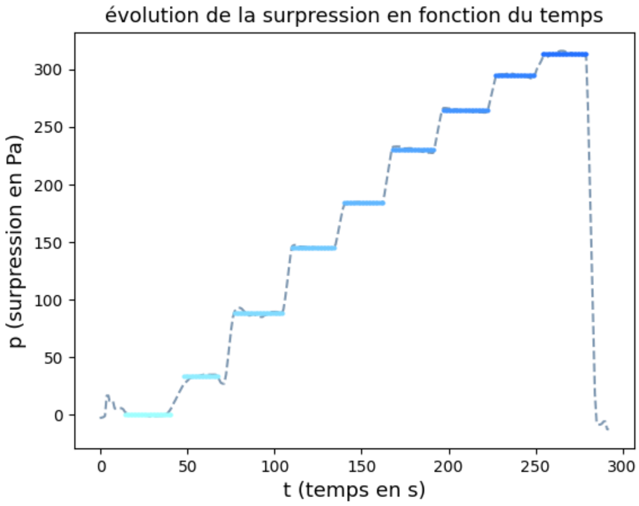


Figure 5: Overpressure recorded across the porous wall as a function of time. The regulator voltage is increased step by step; the Python plateau-detection algorithm automatically identifies the stable levels (highlighted segments) and computes their mean value, here indicated as P_{atm} at rest.

2.4.2 Results

The overpressure component due to the wall is **linear in velocity**:

$$\Delta P_{\text{wall}}(V) = \alpha V + \beta \tag{2.6}$$

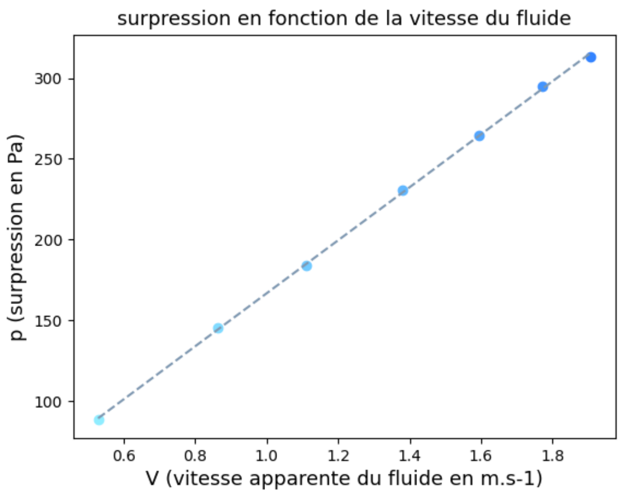


Figure 6: Overpressure due to the porous wall as a function of the apparent fluid velocity V. The dashed line is a linear fit, confirming that $\Delta P_{\text{wall}} \propto V$ (Darcy regime through the fabric). This contribution, reaching up to $\sim 300\,\text{Pa}$ at $V = 1.9\,\text{m}\,\text{s}^{-1}$, is non-negligible and is subtracted from all subsequent bed measurements.

3 Verification of the Laws Governing Fluidization

3.1 Theoretical laws

3.1.1 Darcy's law

Darcy's law describes the laminar flow of a viscous fluid through a porous medium. It establishes a linear relationship between the pressure gradient and the Darcy velocity (or superficial velocity):

Darcy's law

$$\frac{\Delta P}{L} = \frac{\eta}{k} V \tag{3.1}$$

where L is the bed height, η is the dynamic viscosity of the fluid (for air: $\eta \approx 1.8 \times 10^{-5}\,\mathrm{kg\,m^{-1}\,s^{-1}}$), and k is the **permeability** of the medium (m^2).

Microscopic derivation of Darcy's law. We model the porous medium as a set of N capillaries of diameter d and length L_{cap}. For a single capillary, Poiseuille's law gives the parabolic velocity profile:

$$v(r) = \frac{-\Delta P}{4\eta L_{\text{cap}}} \left(r^2 - \frac{d^2}{4} \right) \tag{3.2}$$

The volumetric flow rate in one capillary is:

$$Q_{V,\text{one pore}} = \int_0^{d/2} v(r)\, 2\pi r\, dr = \frac{\pi}{128\eta} \frac{\Delta P}{L_{\text{cap}}} d^4 \tag{3.3}$$

Averaging over the N capillaries and expressing the mean velocity $V = Q_V/S$:

$$V = \frac{Q_V}{S} = N \frac{\pi d^4}{128\eta S} \frac{\Delta P}{L_{\text{cap}}} = \frac{k}{\eta} \frac{\Delta P}{L} \tag{3.4}$$

where $k = N \dfrac{\pi d^4}{128 S} \dfrac{L}{L_{\text{cap}}}$ is the medium permeability, which establishes Darcy's law (3.1).

3.1.2 Ergun's semi-empirical law

At higher velocities, inertial effects become significant. Ergun's law adds a quadratic term in V:

Ergun's law

$$\frac{\Delta P}{L} = \underbrace{\frac{\eta}{k} V}_{\text{Darcy term}} \underbrace{+ 1.8 \frac{\varrho_{\text{fluid}}}{d_p} \frac{\phi}{(1-\phi)^3} V^2}_{\text{inertial term}} \tag{3.5}$$

where:

- $\phi = V_{\text{grains}}/V_{\text{total}}$ is the **solid volume fraction** of the bed,
- d_p is the **mean particle diameter** (m),
- ϱ_{fluid} is the fluid density ($\text{kg}\,\text{m}^{-3}$).

The coefficients can be rewritten as:

$$A(\phi) = \frac{\varrho_{\text{air}}}{d_p} \frac{\phi}{(1-\phi)^3} \tag{3.6}$$

$$B(\phi) = \frac{150}{d_p^2} \frac{\phi^2}{(1-\phi)^3} \eta \tag{3.7}$$

so that: $\dfrac{\partial P}{\partial z}(V, \phi) = A(\phi)V^2 + B(\phi)V$.

3.1.3 Pressure plateau after fluidization

In the fluidized bed regime (beyond V_{mf}), the pressure drop stabilizes. This plateau corresponds to the vertical force balance on all the particles: the fluid pressure thrust exactly balances the apparent weight of the bed. This gives:

Pressure plateau in the fluidized bed

$$\Delta P_{\text{plateau}} = \frac{M_{\text{bed}}\, g}{S} \tag{3.8}$$

where M_{bed} is the total mass of particles in the tube and S is the tube cross-sectional area.

3.2 Experimental protocol

The tube is filled with a mass $M_{\text{bed}} = 400\,\text{g}$ of fine sand (mean radius $r_p = 0.50\,\text{mm}$, density $\varrho_p = 1850\,\text{kg}\,\text{m}^{-3}$). The regulator voltage is increased in successive steps, and the pressure in the pressure cooker is recorded continuously with Phyphox. The Python plateau-detection script extracts the mean value of ΔP for each plateau, and then subtracts the wall component $\Delta P_{\text{wall}}(V)$ measured previously.

3.3 Results

3.3.1 Fluidization plateau

Experimental result — pressure plateau

The measured pressure drop after fluidization is:

$$\Delta P_{\text{exp}} = 1.670 \pm 0.010 \text{ kPa}$$

The theoretical value, calculated from equation (3.8) with $M_{\text{bed}} = 400\,\text{g}$ and $S = \pi(2.75 \text{ cm})^2$, is:

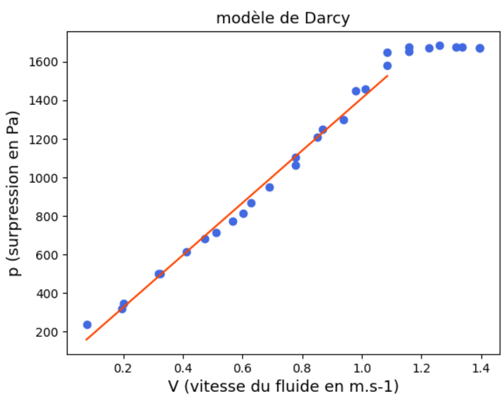
$$\Delta P_{\text{th}} = \frac{0.400 \times 9.81}{\pi(0.0275)^2} = 1.651 \text{ kPa}$$

The normalised deviation $\varepsilon_n = \dfrac{|\Delta P_{\text{exp}} - \Delta P_{\text{th}}|}{\sigma} = 1.7 < 2$ validates the model at the 95% confidence level.

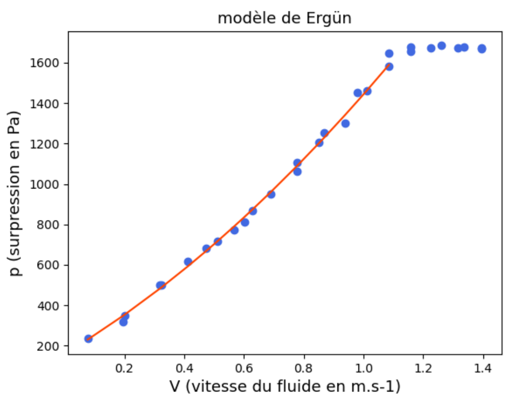
The measured fluidization velocity is $V_{\text{fluidization}} = 1.08\,\text{m}\,\text{s}^{-1}$.

3.3.2 Comparison of Darcy and Ergun models

Figures 7a and 7b show the experimental overpressure data (after subtraction of the wall component) fitted by the Darcy model and the Ergun model respectively. The quality of each fit is assessed through the residuals plotted in Figures 8a and 8b.

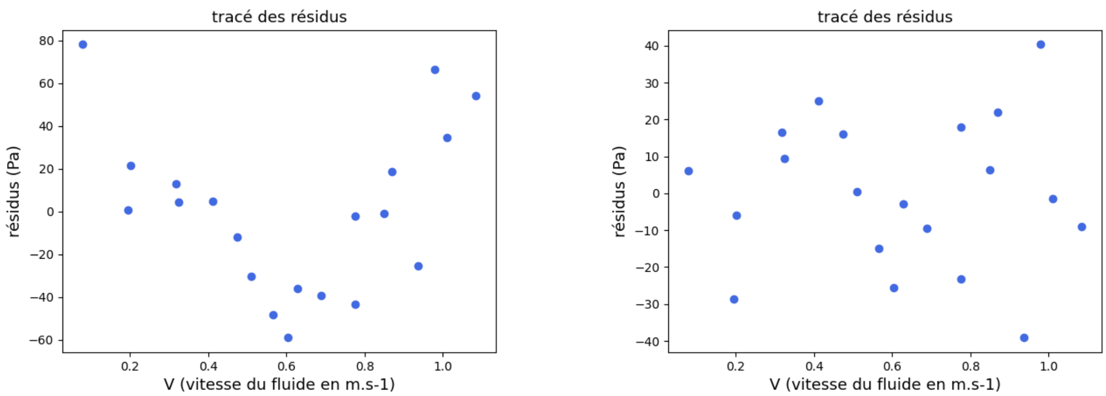


(a) Darcy model (linear fit).



(b) Ergun model (quadratic fit).

Figure 7: Overpressure ΔP as a function of fluid velocity V in the fixed-bed regime, fitted by Darcy's law (left) and Ergun's law (right). Blue dots are experimental measurements after wall correction; the plateau visible at $V \gtrsim 1.1\,\text{m}\,\text{s}^{-1}$ marks the onset of fluidization.



(a) Darcy residuals: a clear U-shaped trend indicates systematic deviation at intermediate velocities.

(b) Ergun residuals: centred around zero with no systematic pattern, confirming the adequacy of the quadratic model.

Figure 8: Residuals (experimental minus model) for the Darcy fit (left) and the Ergun fit (right). The structured non-zero residuals of Darcy's model demonstrate that the inertial term in Ergun's law is necessary at the velocities explored here.

Analysis of the residuals confirms that:

- The **Darcy model** (linear in V) produces residuals that are systematically non-zero and tend to grow with velocity, indicating a bias at higher velocities.
- The **Ergun model** (quadratic in V) produces residuals that are centred around zero and randomly distributed, indicating a better fit and validating the necessity of the inertial term for our velocity range.

These results confirm that Darcy's law alone is insufficient when the particle Reynolds number $Re_p = \varrho_{\text{air}} V d_p / \eta$ is not very small compared to 1.

4 Development of the Numerical Model

4.1 Choice of method: Discrete Element Method (DEM)

The numerical model is based on the **Discrete Element Method (DEM)**, which consists of treating each grain individually by integrating Newton's equations of motion for each particle.

At each time step dt, the computational cycle is as follows:

1. **Computation of forces** on each grain (gravity, inter-grain contact forces, fluid pressure forces, wall contact forces).
2. **Computation of the acceleration** of each grain via Newton's second law.

3. **Integration of the equations of motion** to obtain the new velocities and positions.

4. **Saving the state** of the system and incrementing time.

To simplify the numerical model while preserving the essential phenomena, we study a **2D planar cross-section** of the experimental setup. This dimensional reduction is justified since the fluidization dynamics are primarily vertical.

4.2 Modelling grain–grain collision forces

4.2.1 Soft sphere method

Collisions between grains are modelled by a variant of the **soft sphere method**, drawn from the reference work *Granular Media: Between Fluid and Solid* (CNRS Editions, 2011). In this approach, grains are allowed to slightly overlap, and this overlap generates a restoring force (spring) and a dissipation force (viscous damper).

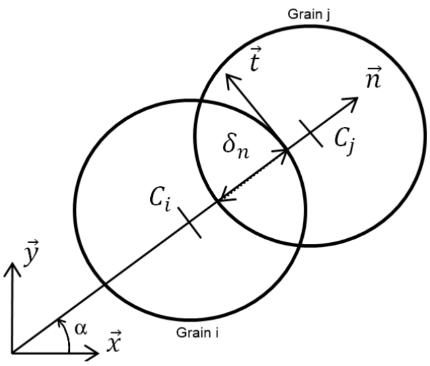


Figure 9: Geometry of a contact between grains i and j. C_i and C_j denote the grain centres; $\vec{n}$ and $\vec{t}$ are the unit vectors normal and tangential to the contact; δ_n is the normal overlap; α is the contact angle with respect to $\vec{x}$. The overlap drives the restoring spring force, while relative velocities along $\vec{n}$ and $\vec{t}$ determine the damping forces.

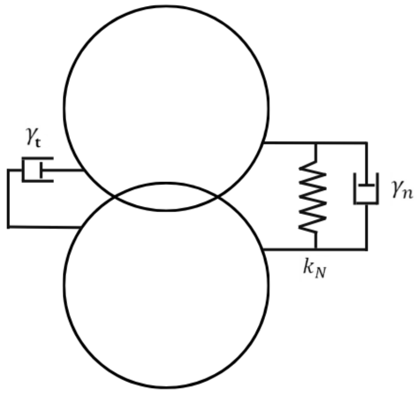


Figure 10: Mechanical analogue of the soft-sphere contact model. In the normal direction, the interaction is a spring (stiffness k_N) in parallel with a viscous dashpot (damping γ_n). In the tangential direction, only a dashpot (γ_t) is applied, modelling dissipation without tangential restoring force.

Let grains i and j have centres C_i and C_j and radius r_p. We define:

- $\vec{n}$ the unit vector from C_i to C_j, with angle $\alpha = \mathrm{atan2}(dy, dx)$.
- $\delta_n = 2r_p - d(C_i, C_j)$ the **normal overlap** (positive if collision).
- v_n and v_t the normal and tangential components of the relative velocity $\vec{v}_{\mathrm{rel}} = \vec{v}_i - \vec{v}_j$.

The normal and tangential forces exerted by j on i are:

$$F_n = -K(2r_p - d) - \gamma v_n \qquad \text{(spring + normal damper)} \tag{4.1}$$

$$F_t = -\gamma v_t \qquad \text{(tangential damper)} \tag{4.2}$$

where K is the **stiffness constant** and γ the **damping coefficient**.

4.2.2 Determination of constants K and γ

These constants are determined from physical considerations on the collision duration. Modelling the collision of two grains as a damped oscillator, Newton's second law along the collision axis gives:

$$\frac{d^2z}{dt^2} + \frac{\gamma}{m}\frac{dz}{dt} + \frac{K_N}{m}(z - r) = -g \tag{4.3}$$

We impose:

- **Low oscillation**: $Q = \dfrac{\sqrt{mK_N}}{\gamma} \simeq 0.5$ (slightly underdamped regime).

- **Collision duration**: $T_{\text{collision}} \simeq 6 \times 10^{-3}\,\text{s}$, which fixes the natural angular frequency $\omega_0 = \dfrac{2\pi}{2T_{\text{collision}}}$.

We then obtain the **orders of magnitude** of the constants:

Collision constants

$$K_N = m\omega_0^2 \qquad \text{and} \qquad \gamma = \frac{m\omega_0}{Q} \tag{4.4}$$

4.2.3 Grain–wall interaction

The interaction of a grain with the rigid wall is modelled by the same spring–damper model, with an additional boundary condition: when a grain is very close to the wall ($y < r_p/2$ or $y > y_{\text{lim}} - r_p/2$, similarly for x), the normal component of its velocity is reversed:

$$v_{y,i} \leftarrow -v_{y,i} \qquad \text{(elastic reflection near the wall)} \tag{4.5}$$

This condition ensures that grains do not leave the simulation domain.

4.3 Efficient collision detection

Naive collision detection (testing every pair of grains) has complexity $O(N^2)$, which is prohibitive for large simulations. We implement here an $O(N)$ algorithm.

4.3.1 Grid partitioning algorithm

The simulation space is divided into square cells of side $2r_p$. At each time step, each grain is assigned to the cell corresponding to its position:

$$(i, j) = \left(\left\lfloor \frac{X}{2r_p} \right\rfloor, \left\lfloor \frac{Y}{2r_p} \right\rfloor \right) \tag{4.6}$$

This partitioning is performed in a single pass over all grains: complexity $O(N)$.

4.3.2 Neighbour search

For each grain, collisions are searched only in the 9 neighbouring cells (current cell and 8 adjacent cells). Since cell size is $2r_p$, two grains can only be in contact if they are in adjacent or the same cell.

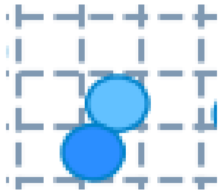


Figure 11: Illustration of the neighbourhood search. The grain under consideration (light blue) can only collide with grains in the 3×3 block of adjacent cells (dashed), limiting the number of pair checks to at most a constant per grain and yielding an overall $O(N)$ complexity.

4.4 Modelling fluid-induced forces

4.4.1 Pressure force on a spherical grain

The fluid exerts a pressure force on each grain. For a spherical grain of radius r immersed in a pressure field $P(z)$ with a gradient along $\vec{e}_z$, the resultant force is computed by integrating the pressure over the grain surface.

Derivation. By axial symmetry, only the component along $\vec{e}_z$ is non-zero. In spherical coordinates centred at $C = (x_0, y_0, z_0)$:

$$\vec{F}_p = \left(\iint_{\text{Grain}} P_{\text{fluid}}\, d\vec{S} \cdot \vec{e}_z \right) \vec{e}_z \tag{4.7}$$

Expanding $P_{\text{fluid}}(z) = -\dfrac{\partial P}{\partial z} z$ (relative pressure) and integrating over the surface $dS = r^2 \sin\theta \, d\theta \, d\varphi$:

$$F_p = \frac{\partial P}{\partial z} \int_0^{2\pi} d\varphi \int_0^{\pi} (z_0 + r\cos\theta)\, r^2 \cos\theta \sin\theta \, d\theta \tag{4.8}$$

$$F_p = 2\pi r^2 \frac{\partial P}{\partial z} \left[z_0 \frac{\cos^2\theta}{2} + r \frac{\cos^3\theta}{3} \right]_0^{\pi} \tag{4.9}$$

The boundary terms give: $\left[\frac{\cos^2\theta}{2}\right]_0^{\pi} = 0$ and $\left[\frac{\cos^3\theta}{3}\right]_0^{\pi} = -\dfrac{2}{3}$.

Fluid pressure force on a grain

$$\vec{F}_p(V, \phi) = \frac{4\pi}{3} r^3 \frac{\partial P}{\partial z}(V, \phi)\, \vec{e}_z \tag{4.10}$$

4.4.2 Local pressure gradient

The pressure gradient is computed locally for each grain using **Ergun's law**, with the volume fraction ϕ in the neighbourhood of the grain:

$$\frac{\partial P}{\partial z}(V, \phi) = A(\phi)V^2 + B(\phi)V \tag{4.11}$$

4.4.3 Computation of the local volume fraction

The volume fraction ϕ in the neighbourhood of grain i is estimated by counting the number of grains $N_{\text{neighbours}}$ in the neighbouring cells (9 cells at most), and relating the volume occupied by these grains to the total volume of the cells:

$$V_g = N_{\text{neighbours}} \times \frac{4}{3}\pi r_p^3 \qquad \text{(grain volume)} \tag{4.12}$$

$$V_0 = N_{\text{cells}} \times (2r_p)^3 \qquad \text{(cell volume)} \tag{4.13}$$

$$\phi = \frac{V_g}{V_0} \tag{4.14}$$

4.5 Integration method: Verlet (Leapfrog) algorithm

4.5.1 Presentation of the method

The integration of the equations of motion is performed using the **Leapfrog Verlet method**, which has symplectic properties (long-term energy conservation) and a truncation error of $O(\Delta t^4)$.

The idea is to stagger the evaluations of position and velocity by half a time step:

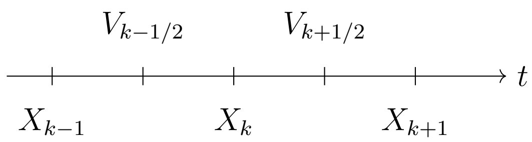


Figure 12: Time diagram of the Leapfrog Verlet method. Positions X_k are evaluated at integer steps, velocities $V_{k\pm 1/2}$ at half-integer steps.

4.5.2 Derivation of the scheme

Expanding $X(t \pm \Delta t)$ as a Taylor series to order 4:

$$X(t - \Delta t) = X(t) - v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 - \frac{1}{6}\dddot{X}\Delta t^3 + O(\Delta t^4) \tag{4.15}$$

$$X(t + \Delta t) = X(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 + \frac{1}{6}\dddot{X}\Delta t^3 + O(\Delta t^4) \tag{4.16}$$

Setting:

$$V_{k-1/2} \equiv v(t)\Delta t - \frac{1}{2}a(t)\Delta t^2 \tag{4.17}$$

$$V_{k+1/2} \equiv v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 \tag{4.18}$$

We obtain Verlet's formula:

Leapfrog Verlet algorithm

$$X(t+\Delta t) = 2X(t) - X(t-\Delta t) + a(t)\Delta t^2 + O(\Delta t^4) \tag{4.19}$$

The truncation error is $O(\Delta t^4)$, which is significantly better than Euler integration ($O(\Delta t^2)$). Moreover, Verlet is symplectic: it conserves the energy of the system over long simulations, unlike Euler which drifts.

In practice, the leapfrog implementation uses half-steps:

$$V_{x,i}^{k+1/2} = V_{x,i}^{k} + \frac{1}{2}\Delta t\, A_{x,i}^{k} \tag{4.20}$$

$$X_{i}^{k+1} = X_{i}^{k} + \Delta t\, V_{x,i}^{k+1/2} \tag{4.21}$$

$$A_{x,i}^{k+1} = F_{x,i}^{k+1}/m \quad \text{(forces recomputed at new positions)} \tag{4.22}$$

$$V_{x,i}^{k+1} = V_{x,i}^{k+1/2} + \frac{1}{2}\Delta t\, A_{x,i}^{k+1} \tag{4.23}$$

The chosen time step is $\Delta t = 3 \times 10^{-4}\,\text{s}$, well below the collision time $T_{\text{collision}} \simeq 6 \times 10^{-3}\,\text{s}$, ensuring numerical stability.

4.6 Complete description of the model

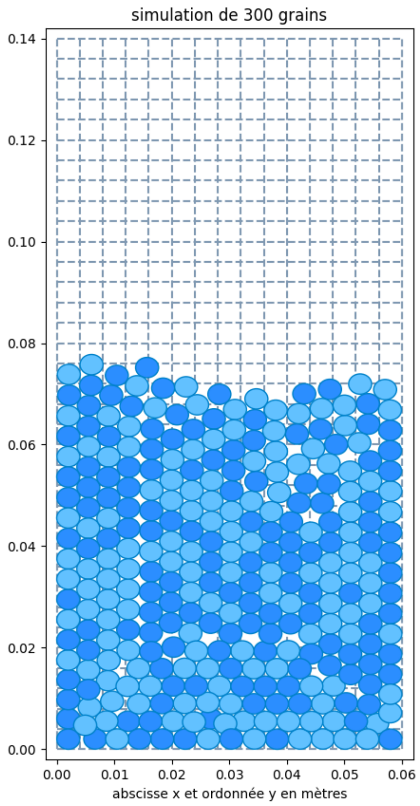
The complete algorithm can be summarised as follows:

Algorithm 1 DEM simulation of a fluidized bed

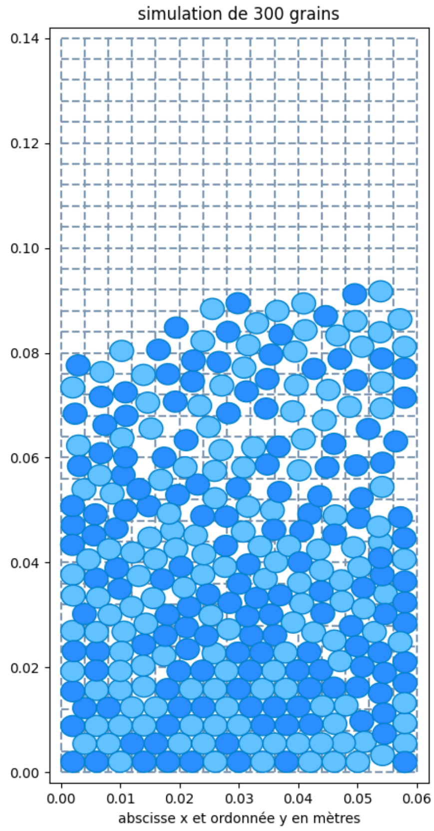
Initialisation: random positions (X_i, Y_i), zero velocities, $t = 0$, $V_{\text{fluid}} = 0$
while $t < t_{\max}$ **do**
    Distribute the N grains in the 2D grid in $O(N)$
    for each grain i **do**
        Search for neighbours in the 9 adjacent cells
        for each neighbour j in contact with i **do**
            Compute (F_n, F_t, α) via the soft sphere model
            Project and accumulate into $\vec{F}_i$ and $\vec{F}_j$
        end for
        Compute ϕ_i (local volume fraction)
        Compute ∇P_i via Ergun's law
        Accumulate the fluid pressure force $\vec{F}_{p,i}$ (equation 4.10)
        Apply wall boundary conditions to $(V_{x,i}, V_{y,i})$
    end for
    Integrate all positions/velocities by Leapfrog Verlet (equation 4.19)
    $t \leftarrow t + \Delta t$, $V_{\text{fluid}} \leftarrow V_{\text{fluid}} + dV$
end while

4.7 First visual results

The simulation is run with $N = 300$ grains. Figure 13a shows the compact packing obtained at rest ($V_{\text{fluid}} = 0$), while Figure 13b shows the expanded, agitated state reached when $V_{\text{fluid}} > 1.5\,\mathrm{m\,s^{-1}}$. This behaviour is qualitatively consistent with the experimental observations: the bed height roughly increases by a factor of 1.5 upon fluidization.



(a) Fixed bed ($V_{\text{fluid}} = 0$): grains settle into a compact random packing occupying roughly the lower half of the simulation domain.



(b) Fluidized bed ($V_{\text{fluid}} > 1.5\,\mathrm{m\,s^{-1}}$): grains are dispersed throughout the domain, with significant expansion and visible void regions.

Figure 13: Snapshots of the DEM simulation with $N = 300$ grains. The dashed grid shows the cell partition used for $O(N)$ collision detection. Grain colour distinguishes the two random colours used for visualisation.

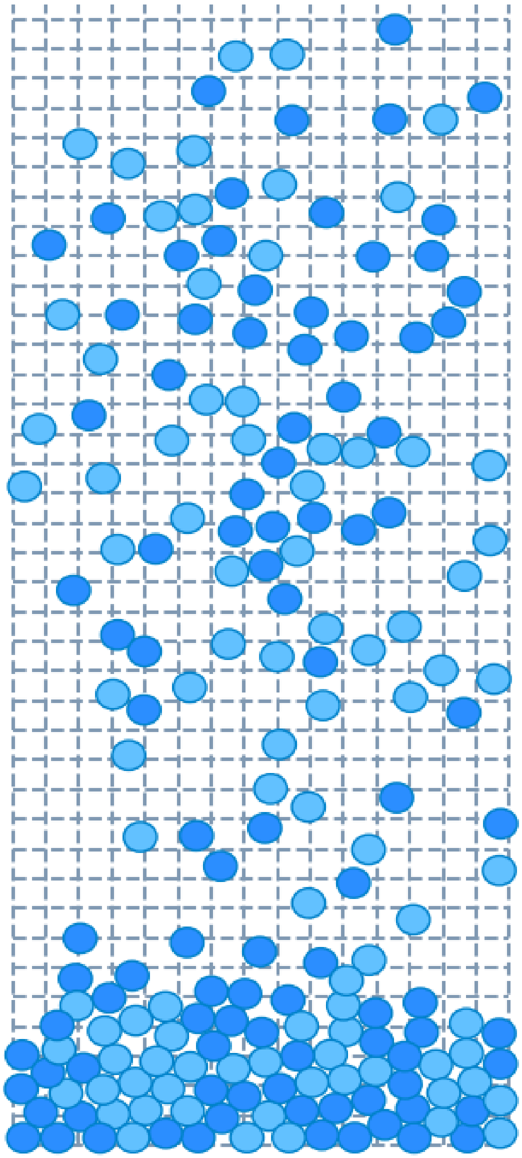


Figure 14: Full view of the DEM simulation beyond the fluidization regime. The grain density decreases progressively from the bottom (dense packing) to the top (sparse dispersion), reproducing qualitatively the spatial structure observed experimentally.

5 Validation of the Numerical Model

5.1 Validation strategy

To validate the model, the fluidization velocities predicted numerically are compared to those measured experimentally, for **five granular media** with different characteristics.

5.1.1 Granular media studied

The five media are:

Table 1: Characteristics of the granular media studied

Medium	Mean radius (mm)	ϱ_p ($\mathrm{kg\,m^{-3}}$)	$V_{\mathrm{mf,exp}}$ ($\mathrm{m\,s^{-1}}$)
Fine sand	0.50	1850	1.08
Salt	1.50	848	0.85
Chia seeds	0.65	853	0.64
Quinoa	0.90	1457	0.63
Lentils	1.75	1547	0.80

The physical characteristics (mean radius, density) are measured experimentally:

- The radius is measured with a **micrometer** (precision $\pm 0.01\,\mathrm{mm}$).
- For insoluble materials (sand, chia seeds, quinoa, lentils), density was estimated via volume displacement. For soluble materials (salt), density values were taken from literature.

5.2 Numerical determination of V_{mf}

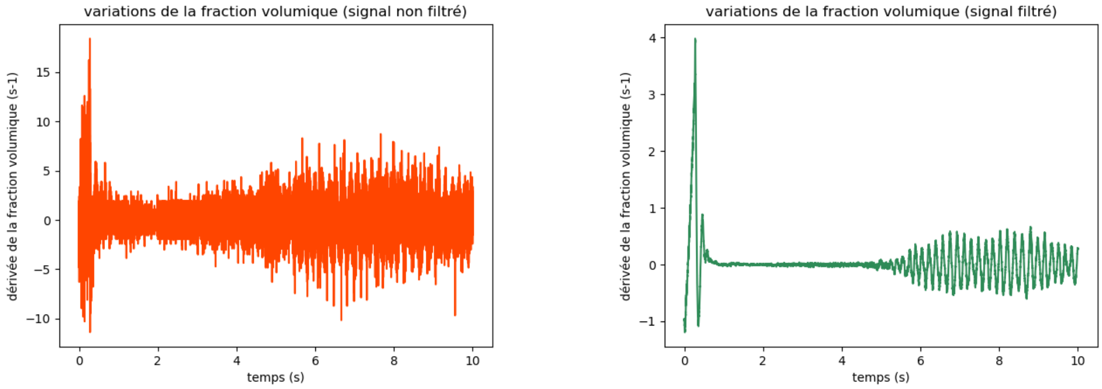
In the simulation, the fluid velocity V is increased linearly with time. The mean volume fraction $\phi_M = \langle \phi_i \rangle$ is recorded at each time step. Fluidization corresponds to an **abrupt transition**: the volume fraction oscillates when the bed lifts.

The time derivative $d\phi_M/dt$ is therefore computed; it presents oscillations centered on the x-axis at the moment of fluidization. The numerical fluidization velocity $V_{\mathrm{mf,num}}$ is read at the instant corresponding to this transition.

5.2.1 Signal filtering

The raw derivative $d\phi_M/dt$ is very noisy. This agitation arises from the finite time step Δt: since Δt is not infinitesimally small, the discrete nature of the simulation introduces high-frequency fluctuations in the grain dynamics. A first-order low-pass filter with time constant $\tau = 2.5 \times 10^{-2}\,\mathrm{s}$ is applied:

$$\left(\frac{d\phi}{dt}\right)_{\text{filtered}}[n] = \left(\frac{d\phi}{dt}\right)_{\text{filtered}}[n-1] + \frac{\Delta t}{\tau}\left(\frac{d\phi}{dt}[n-1] - \left(\frac{d\phi}{dt}\right)_{\text{filtered}}[n-1]\right) \tag{5.1}$$



(a) Raw (unfiltered) derivative $d\phi/dt$: the signal is dominated by high-frequency noise from grain collisions, making it impossible to identify the fluidization transition directly.

(b) Filtered derivative $d\phi/dt$: the low-pass filter ($\tau = 2.5 \times 10^{-2}$ s) removes the collision noise while preserving the fluidization transition.

Figure 15: Time derivative of the mean volume fraction ϕ_M before (left) and after (right) low-pass filtering. The fluid velocity V increases monotonically from left to right.

5.3 Results and comparison

Figure 16 compares the numerical fluidization velocities $V_{\mathrm{mf,num}}$ with the experimental values $V_{\mathrm{mf,exp}}$ for all five granular media.

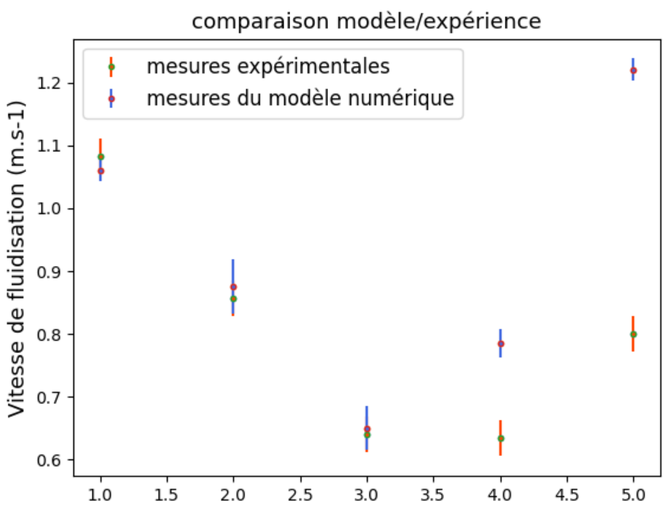


Figure 16: Comparison of the fluidization velocities V_{mf} obtained experimentally (orange error bars) and from the DEM numerical model (blue error bars) for the five granular media studied (1: fine sand, 2: salt, 3: chia seeds, 4: quinoa, 5: lentils). Error bars represent the standard deviation of repeated measurements.

Comparison: numerical model vs. experiment

Error bars systematically overlap for fine sand, salt, and chia seeds (media 1 to 3). For quinoa and lentils (media 4 and 5), the error bars do not overlap due to larger deviations.

5.4 Limitations of the model

Two physical effects are not accounted for in the current model:

1. **Mixture of grains of different sizes:** In reality, granular media are not perfectly monodisperse. The presence of grains of varying sizes creates a *more compact packing* (higher volume fraction), modifying the fluidization velocity.

2. **Non-sphericity of grains:** The model assumes spherical particles. However, irregularly shaped grains (e.g. lentils (oval) and quinoa (flat)) alter the fluid field lines within the packing, affecting the drag coefficient and hence the actual pressure gradient. This partly explains the deviations observed for lentils and quinoa.

6 General Discussion

6.1 Assessment of the experimental approach

The experimental approach, although constrained by the available laboratory resources, proved robust. Using the Phyphox app as a precision pressure sensor is an effective solution, achieving pascal-level accuracy with consumer-grade hardware. Careful calibration of the flow rate–voltage characteristic and subtraction of the porous wall contribution enabled reliable measurements to be obtained.

6.2 Assessment of the numerical model

The DEM model developed captures the essential phenomena:

- The phase transition from fixed bed to fluidized bed.
- The dependence of V_{mf} on the physical properties of the grains (radius, density).
- Bed expansion beyond V_{mf}.

However, it does not account for the geometry of the media, which has resulted in less reliable results for quinoa and lentils.

The $O(N)$ collision detection algorithm makes the model suitable for larger-scale simulations. The Verlet integration method guarantees long-term energy stability.

6.3 Perspectives

Several improvements could enrich this work:

1. **3D extension:** Moving from a 2D to a 3D simulation would better capture the actual geometry of the experimental setup.
2. **Size distribution:** Introducing a log-normal distribution of radii to model polydisperse media.
3. **Grain shape:** Using super-ellipsoids or convex polyhedra to model non-spherical grains.
4. **Bubbling regime:** At very high velocities, the fluidized bed enters a *bubbling* regime, with the formation of macroscopic gas pockets. This regime is not captured by the current model.

7 Conclusion

This work has enabled a comprehensive experimental and numerical study of the fluidization of granular media. The main results are:

- The development of an original and precise experimental apparatus, using a smartphone pressure sensor (Phyphox app) to measure overpressures with pascal-level resolution.
- Experimental validation of Darcy's and Ergun's laws for flow through porous media, confirming that Darcy's law alone is insufficient for the velocities of interest (presence of a quadratic inertial term).
- Confirmation of the relation $\Delta P_{\text{plateau}} = M_{\text{bed}} g/S$ in the fluidized regime, with a normalised deviation of 1.7 (below the threshold of 2).
- Development of a complete DEM numerical model (collision forces, fluid pressure forces via Ergun, Verlet integration, $O(N)$ collision detection).
- Validation of the model on three spherical granular media, with accurate results between numerical predictions and experimental measurements.

This project perfectly illustrates the richness of the theme *Transformation, transition, conversion*: fluidization is a genuine mechanical phase transition, with all the physical richness and application diversity that entails.

A Appendix: Summary of Physical Parameters

Table 2: Physical parameters used in the simulations

Parameter	**Value**	**Description**
η	1.8×10^{-5} kg m^{-1} s^{-1}	Dynamic viscosity of air
ϱ_{air}	1.2 kg m^{-3}	Density of air
M_{air}	29×10^{-3} kg mol^{-1}	Molar mass of air
g	9.81 m s^{-2}	Gravitational acceleration
T	298 K	Ambient temperature
V_{cooker}	5.5×10^{-3} m^3	Volume of the pressure cooker
D_{tube}	5.5×10^{-2} m	Inner diameter of the tube
S	$\pi(2.75)^2 \times 10^{-4}$ m^2	Cross-sectional area of tube
N	300	Number of grains in simulation
Δt	3×10^{-4} s	Integration time step
Q	0.65	Collision quality factor
$T_{\text{collision}}$	6×10^{-3} s	Typical collision duration
A	150	Kozeny-Carman coefficient

B Appendix: Kozeny–Carman Law and Permeability

The permeability k of the particle bed is calculated from the Kozeny–Carman relation:

$$k(\phi) = \frac{(2r_p)^2(1-\phi)^3}{A\phi^2} \tag{B.1}$$

where $A = 150$ is the Kozeny–Carman constant. This relation is a semi-empirical approximation valid for porous media composed of spheres.

C Appendix: Complete Flowchart of the Numerical Model

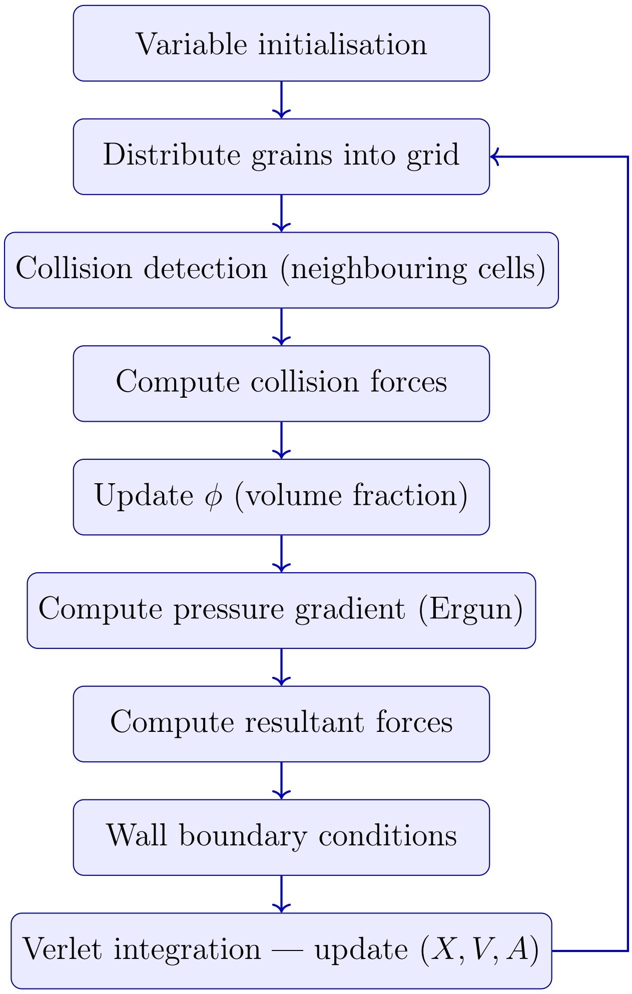


Figure 17: Complete flowchart of the DEM numerical model. The main loop (return arrow) is executed at each time step Δt.

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